Module-4: First Law of Thermodynamics

Thermodynamics (NMEC103) : (Winter 2024-25)

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If we express internal energy and enthalpy as follows:

$$u = f(T,v)\,;\; h = f(T,p)$$

$$du = \left(\frac{\partial u}{\partial T}\right)_v dT + \left(\frac{\partial u}{\partial v}\right)_T dv$$

$$dh = \left(\frac{\partial h}{\partial T}\right)_p dT + \left(\frac{\partial u}{\partial p}\right)_T dp$$

Definition of c_p and c_v

$$c_v = \left(\frac{\partial u}{\partial T}\right)_v$$

$$c_p = \left(\frac{\partial h}{\partial T}\right)_p$$

Hence:

$$du = c_v dT + \left(\frac{\partial u}{\partial v}\right)_T dv$$

$$dh = c_p dT + \left(\frac{\partial u}{\partial p}\right)_T dp$$

Important : These two quantities (c_p & c_v) are properties of any phase of a pure substance . For a two phase or three phase mixture, p and T are not independent, so c_p and c_v are not evaluated

$$du = c_v dT + \left(\frac{\partial u}{\partial v}\right)_T dv$$

$$dh = c_p dT + \left(\frac{\partial u}{\partial p}\right)_T dp$$

From above analysis it can be concluded that for any pure substance:

Important

For *constant volume process:* $du = c_v dT$

For constant pressure process: $dh = c_p dT$

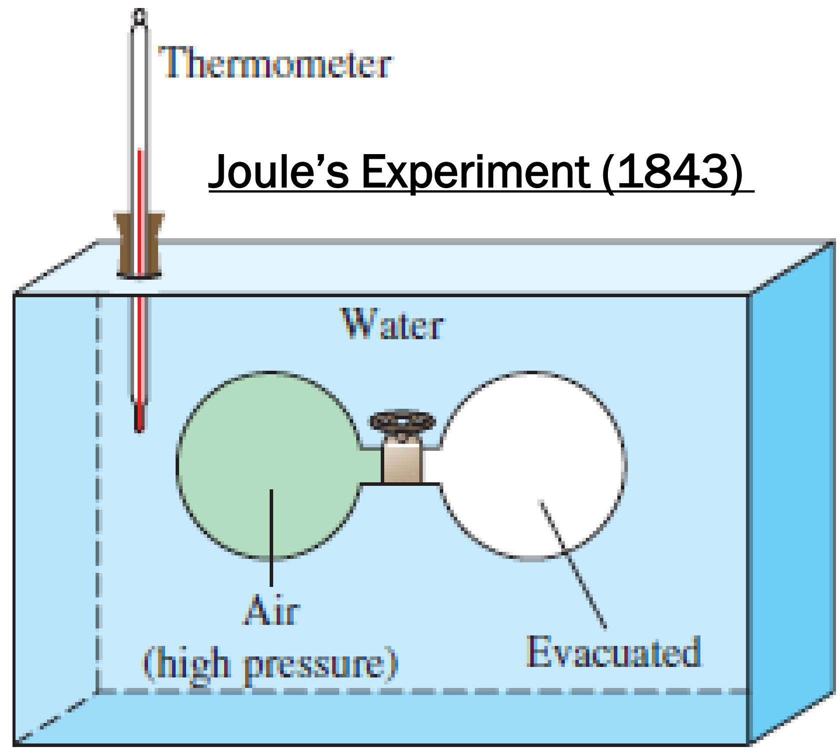
$u, h, C_v, \& C_p$ of ideal gases

$$Pv = RT \rightarrow Ideal\ Gas$$

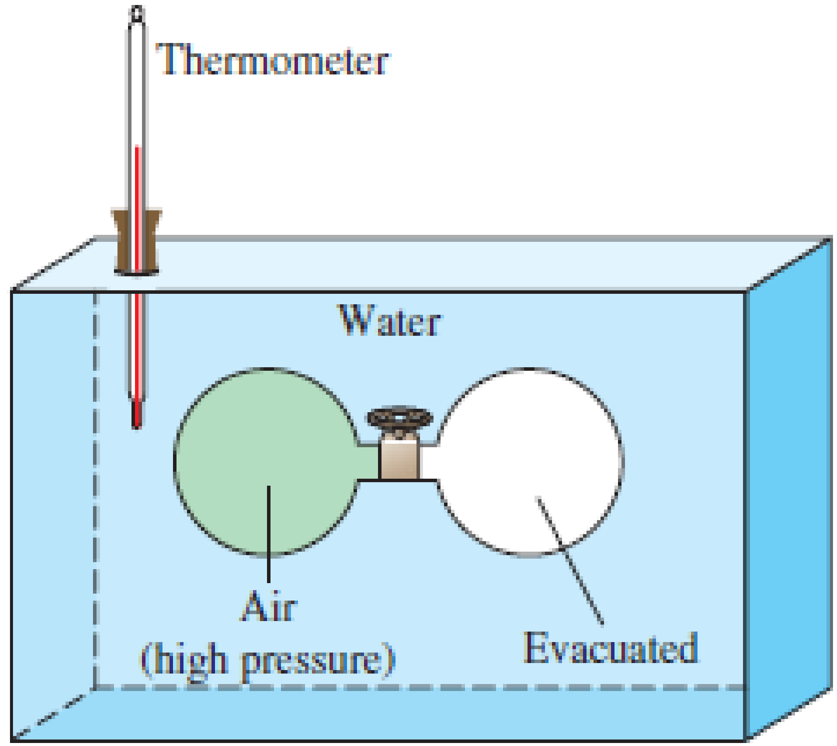
For an ideal gas it can be shown th

$$u = f(T)$$

Initially vessel A contains air and vessel B is highly evacuated. After thermal equilibrium , the valve V is opened allowing the pressure in vessel A & B to equalize. No change in thermometer of the water bath was observed during or after the process.



Because there is no change in temperature Joule concluded that there was no heat transfer between water and air. Since the work is also zero (Free expansion), it was concluded from the 1st law that there was no change in internal energy of the gas. Since P and v changes during the process, it was concluded that $u \neq f(p\&v) \Longrightarrow dU = 0 \; as \; \delta Q = 0; \delta W = 0$



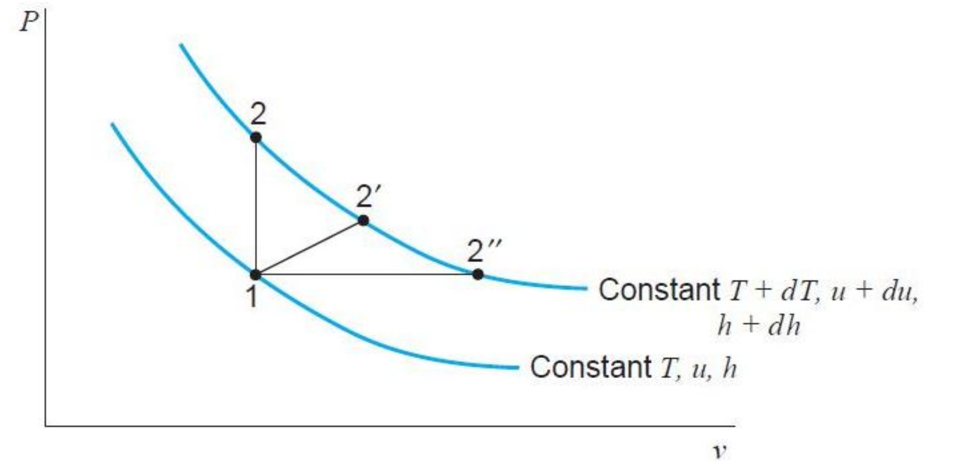
Note: Since air is not exactly an ideal gas, a small change in temperature will be observed when very accurate measurement are made in Joule's experiment. It appears Joule repeated this experiment with real gases and found $u = f(p, v, T)$

We will see in later chapters that 1st and 2nd law together with the help of ideal gas EOS show that the internal energy and enthalpy are function of temperature only. Before it was recognised that this conclusion could be deduced from the principle of thermodynamics, it had been reached from experiments first performed by Joule

Zero Pressure Specific Heats

$$C_{v0} = \left(\frac{\partial u}{\partial T}\right)_v = \frac{du}{dT}$$

$$C_{p0} = \left(\frac{\partial h}{\partial T}\right)_p = \frac{dh}{dT}$$



From state 1 the high temperature can be reached by variety of paths (e.g. 1-2, 1-2', 1-2''...) and in each case the final state is different. However, regardless of the path, the change in the internal energy and enthalpy are the same.

Hence for an ideal gas like internal energy and enthalpy $c_{p,o}$ and $c_{v,0}$ are functions of temperature only $C_{v,0} = f(T)$ $\quad C_{p,0} = f(T)$

So for an ideal gas for any process: $du = C_{v,0} dT \quad dh = C_{p,0} dT$

For an ideal gas: $C_{p,0} - C_{v,0} = R = Constant \quad \frac{C_{p,0}}{C_{v,0}} = \gamma = f(T)$

$$C_{v,0} = \frac{R}{\gamma - 1} \qquad C_{p,0} = \frac{\gamma R}{\gamma - 1}$$

- "Statistical Mechanics" equipartition of energy principle states that when only translational, rotational, and vibrational degrees of freedom are considered , the total energy of molecule is divided equally among there degrees of freedom
- The internal energy per degree of freedom (translational, rotational or vibrational motion of a molecule) in a gas is equal to $\frac{1}{2}\kappa T$ under equilibrium conditions. Here $\kappa = \frac{\bar{R}}{N_{AV}}$
- If the number of translational, rotational and vibrational degrees of freedom per molecule is equal to "D" then the internal energy per molecule $= D\frac{1}{2}\kappa T$

For one kilomole (kmol) of the gas, the molar internal energy is given as)

$$\bar{u} = N_{AV} D \frac{\kappa T}{2} = D \frac{\bar{R}T}{2}$$

$$\bar{h} = \bar{u} + p\bar{v} = D \frac{\bar{R}T}{2} + \bar{R}T = \left(\frac{D+2}{2}\right) \bar{R}T$$

$$\bar{C}_{v,o} = D \frac{\bar{R}}{2}$$

$$\bar{C}_{p,o} = (D+2) \frac{\bar{R}}{2}$$

$$u = \frac{\bar{u}}{M} = D \frac{RT}{2}$$

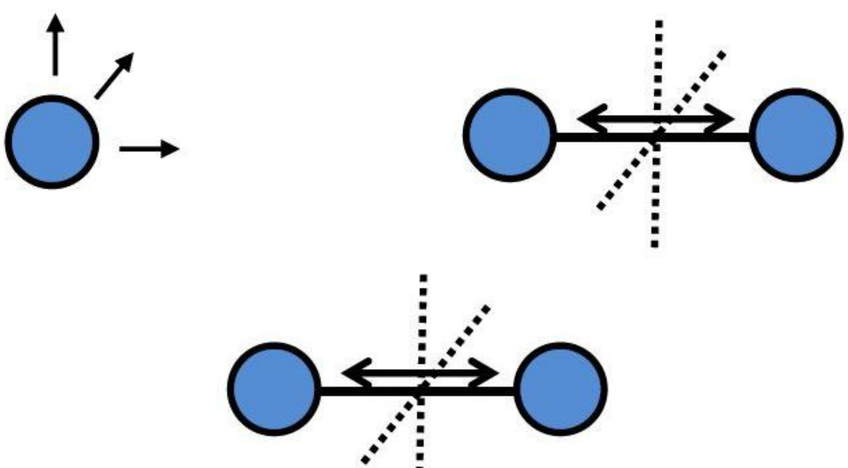
$$h = \frac{\bar{h}}{M} = (D+2) \frac{RT}{2}$$

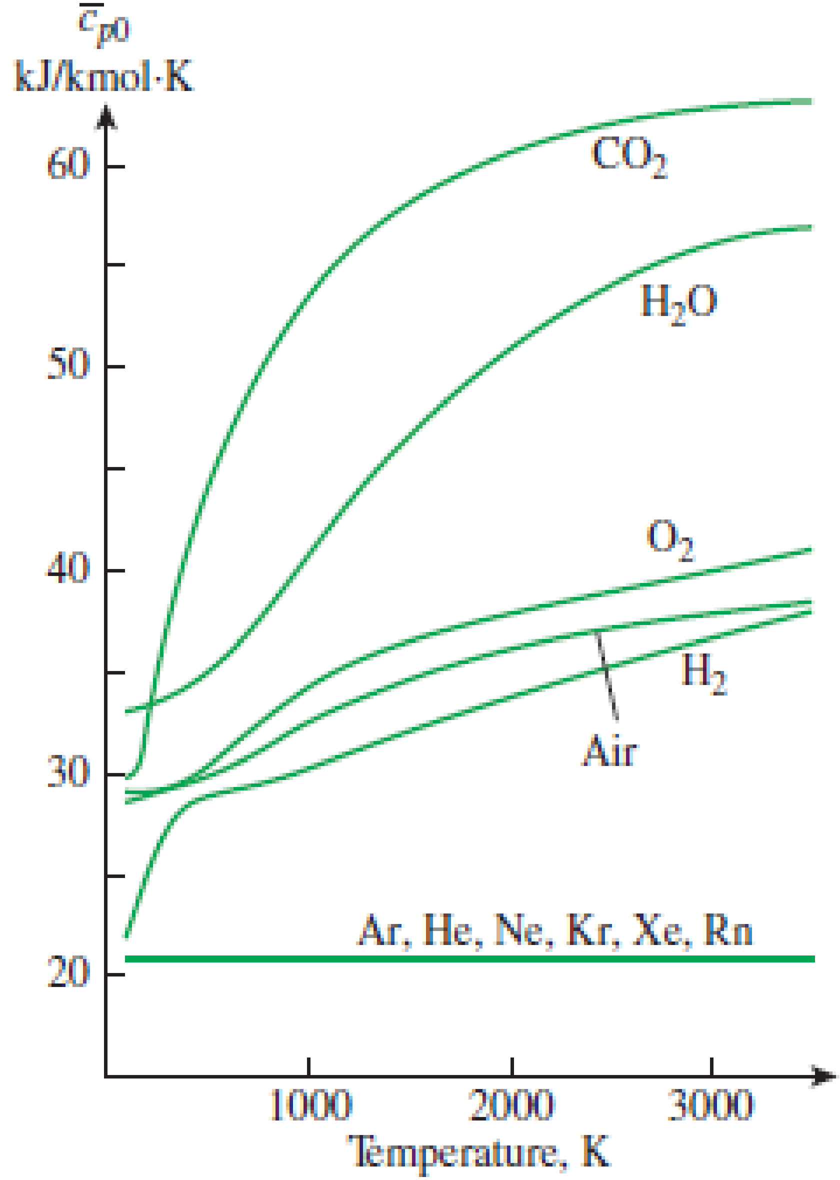
$$C_{v,o} = D \frac{R}{2}$$

$$C_{p,o} = (D+2) \frac{R}{2}$$

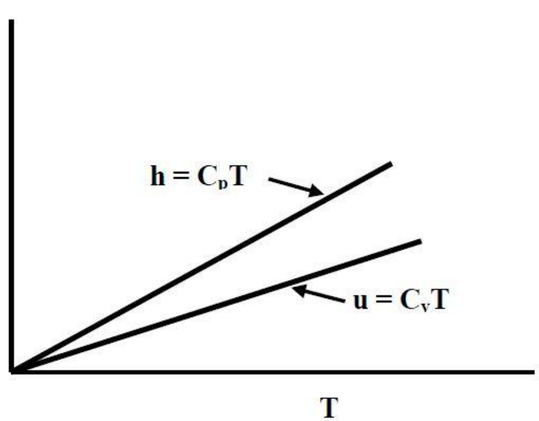
$$\gamma = \frac{(D+2)}{D}$$

1. For a diatomic gas such as nitrogen or oxygen at low temperatures (when vibrational mode is not excited) D = 5, due to 3 translational degrees of freedom (linear motion in x,y and z directions) and 2 rotational degrees of freedom. For monoatomic only translational (D=3)

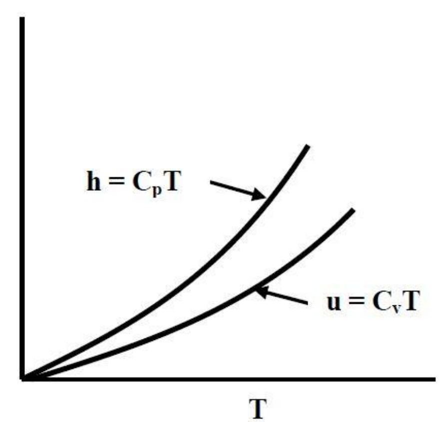




1. The number of degrees of freedom may change with temperature because the vibrational mode gets excited at high temperatures. As temperature increases, the fraction of molecules with vibrational excitation will increase and this results in D (and so γ as well as $C_p$and C_v) varying as functions of temperature.
2. Ideal gases can be further classified into Calorifically Perfect Gases (or simply perfect gases) and Calorifically Semi- Perfect Gases. For perfect gases, the specific heats are constants and hence, the specific internal energy and specific enthalpy are linear functions of temperature. For semi-perfect gases, both u and h increase as non-linear functions of temperature



Calorifically Perfect Gas



Calorifically Semi-Perfect Gas

Frictionless Quasiequilibrium Processes of Ideal Gas with Constant Specific Heat

In case of a simple compressible closed system undergoing a frictionless quasi-static process, the work interaction is given by:

$$W = \int_1^2 p dV$$

Thus, for a constant-volume process $dV = 0$, the quasi-static work is zero and the heat interaction is found by applying 1st law to the closed stationary simple system

$$\boldsymbol{Q = \Delta U + W = \Delta U = m\int_1^2 c_v dT}$$

For constant c_v $\quad \boldsymbol{Q = mc_v(T_2 - T_1)}$

For a constant pressure process, $\boldsymbol{dp = 0}$

$$\boldsymbol{W = \int_1^2 p dV = p(V_2 - V_1) = mR(T_2 - T_1)}$$

$$Q = \Delta U + W = \Delta U + p(V_2 - V_1) = \Delta H = \int_1^2 mc_p dT$$

For constant c_p $$Q = mc_p(T_2 - T_1)$$

For ideal gas with constant c_p & c_v in a closed system undergoing constant pressure process

$$Q = \Delta U + W = mc_v(T_2 - T_1) + p(V_2 - V_1) = mc_v(T_2 - T_1) + mR(T_2 - T_1) = m(c_v + R)(T_2 - T_1) = m\,c_p\,(T_2 - T_1)$$

For an isothermal process in an ideal gas

$$\Delta U = \Delta H = 0$$

$$Q = W = \int_1^2 pdV = mRT ln\frac{V_2}{V_1} = p_1 V_1 ln\frac{p_1}{p_2} = RT_1\, ln\frac{p_1}{p_2}$$

Frictionless Quasiequilibrium Adiabatic Process of Ideal Gas with Constant Specific Heat

For an adiabatic process $\delta Q = 0$, and the 1st law of thermodynamics in differential form is

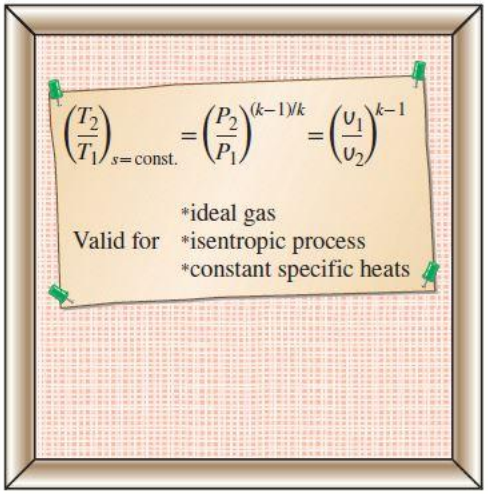
$$dU = -\delta W$$

For an ideal gas undergoing quasi-static friction less adiabatic process

$$mc_v dT = -pdV$$

$$\frac{mc_v dT}{mRT} = \frac{-pdV}{pV}$$

$$\frac{c_v}{R}\frac{dT}{T} = \frac{-dV}{V}$$



Integrating above equation for constant c_v

$$\frac{c_v}{R} ln\frac{T_2}{T_1} = -ln\frac{V_2}{V_1} \qquad c_v = \frac{R}{\gamma - 1} \qquad \frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$$

Displacement work during a Frictionless Quasiequilibrium Adiabatic Process of Calorifically Perfect Gas

$$W = -\Delta U = mc_v(T_1 - T_2)$$

$$W = \frac{1}{\gamma - 1}(p_1V_1 - p_2V_2)$$

$$W = \frac{p_1V_1}{\gamma - 1}\left(1 - \frac{p_2V_2}{p_1V_1}\right)$$

$$W = \frac{mRT_1}{\gamma - 1}\left(1 - \frac{p_2}{p_1}\left(\frac{p_2}{p_1}\right)^{-\left(\frac{1}{\gamma}\right)}\right)$$

$$W = \frac{mRT_1}{\gamma - 1}\left(1 - \left(\frac{p_2}{p_1}\right)^{(\gamma-1)/\gamma}\right)$$

Polytropic Process

$$\frac{T}{T_1} = \left(\frac{V}{V_1}\right)^{1-k} = \left(\frac{p}{p_1}\right)^{k-1/k}$$

Actual processes of ideal gas compression and expansion are neither adiabatic nor isothermal . Actually, these processes can be fairly well approximated as by polytropic processes with $1 < k < \gamma$

$$W = \int_1^2 p dV = p_1 V_1^k \int_1^2 \frac{dV}{V^k} = \frac{p_1 V_1^k}{1-k}\left(V_2^{1-k} - V_1^{1-k}\right)$$

$$W = \frac{p_1 V_1^k}{1-k}\left(V_2^{1-k} - V_1^{1-k}\right) = \frac{1}{1-k}\left(p_2 V_2^k V_2^{1-k} - p_1 V_1^k V_1^{1-k}\right)$$

$$W = \frac{1}{1-k}\left(p_2 V_2 - p_1 V_1\right) = \frac{mR}{k-1}\left(T_1 - T_2\right)$$

$$W = \frac{mRT_1}{k-1}\left(1 - \left(\frac{T_2}{T_1}\right)\right) = \frac{mRT_1}{k-1}\left(1 - \left(\frac{p_2}{p_1}\right)^{(k-1)/k}\right)$$

The heat interaction is calculated from 1st law

$$Q = \Delta U + W = mc_v(T_2 - T_1) + \frac{mR}{k-1}(T_1 - T_2)$$

$$Q = \Delta U + W = \frac{mR}{\gamma - 1}(T_2 - T_1) + \frac{mR}{k-1}(T_1 - T_2)$$

$$Q = \Delta U + W = mR(T_2 - T_1)\left[\frac{1}{\gamma - 1} - \frac{1}{k-1}\right]$$

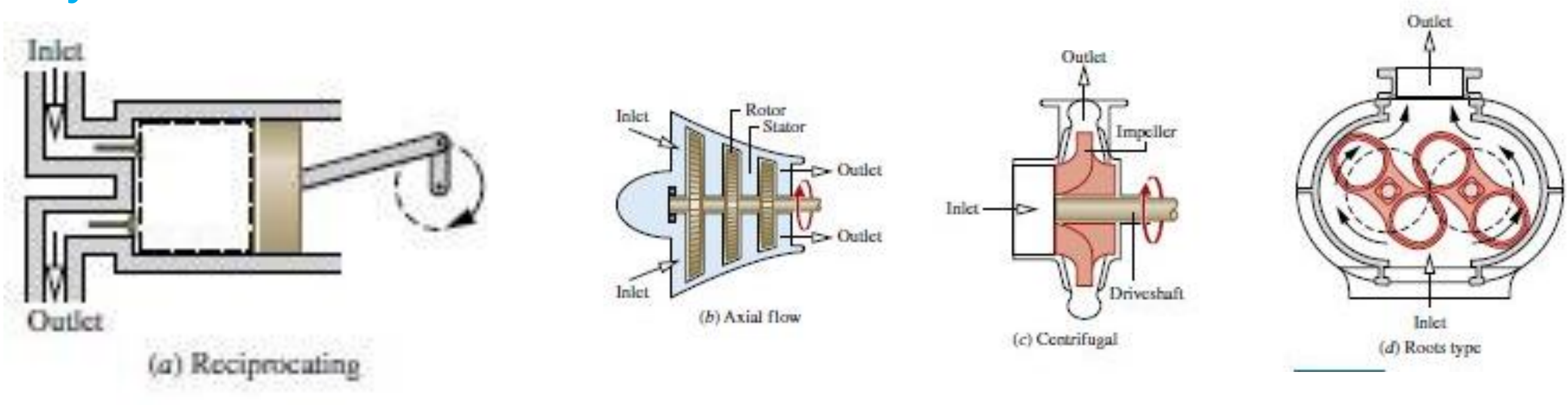
$$Q = mR\left[\frac{k-\gamma}{(\gamma-1)(k-1)}\right](T_2 - T_1)$$

$$Q = m\left\{c_v\left[\frac{(k-\gamma)}{(k-1)}\right]\right\}(T_2 - T_1)$$

$$Q = mc_n(T_2 - T_1)$$

1^{st} Law for Control Volume

In Chapter 1, **'Control Volume'** was defined as a specified region on which attention is focused. Apart from energy interactions in the form of heat or work with the surroundings, it can also admit in-flow (m_i) or out-flow of mass (m_e) through inlet or exit, respectively. Practical devices involving the flow of a fluid, such as a turbine, pump, compressor, nozzle, valve, boiler, condenser, etc. can be modeled as CV. The form of 1^{st} Law for a Control Volume is different from that of a system, on account of the fact that there is mass exchange. Moreover, in addition to the internal energy, forms of energy storage such as the kinetic energy and potential energy could be important in many cases.



(a) Reciprocating
(b) Axial flow
(c) Centrifugal
(d) Roots type

First law applied to a control volume

Unlike a closed system a control volume is permeable for both energy (heat & work) and mass interactions. So before applying energy balance it is necessary to apply conservation of mass.

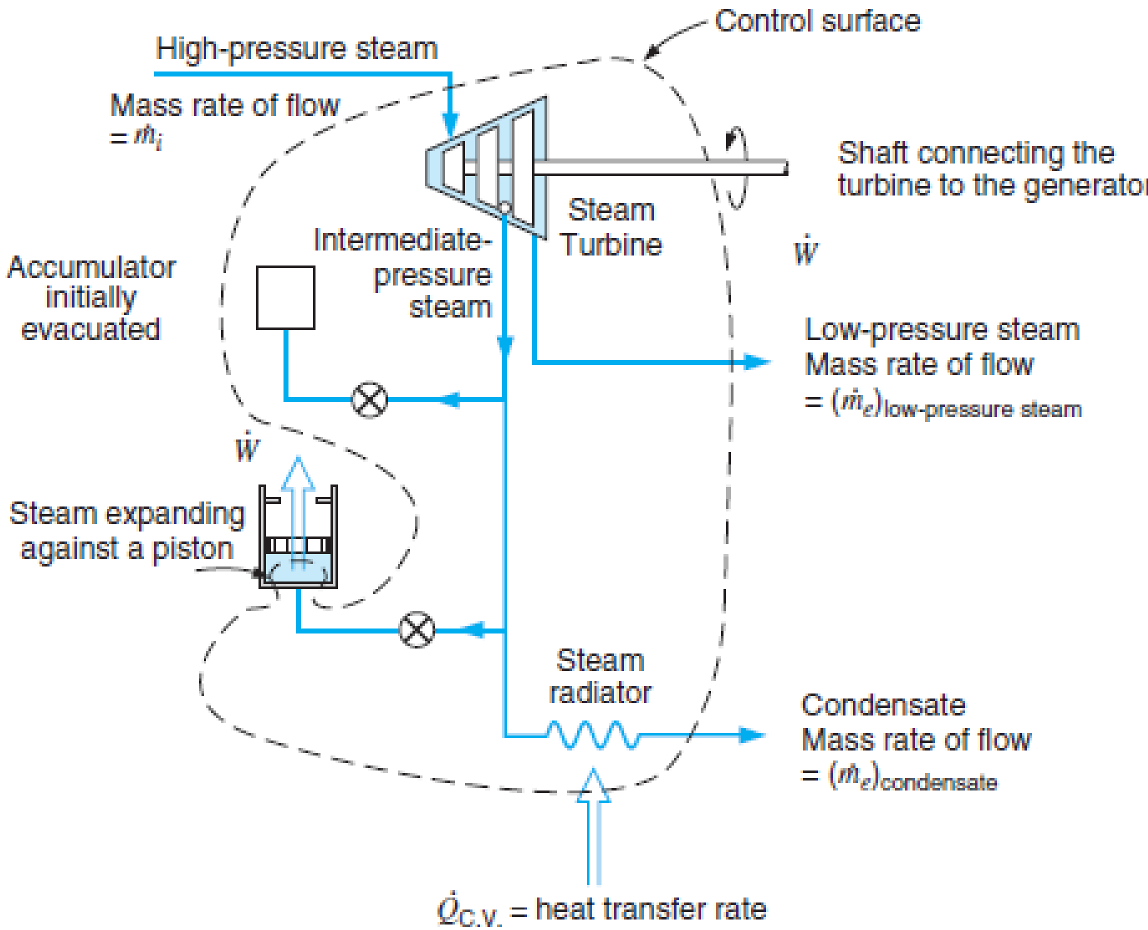
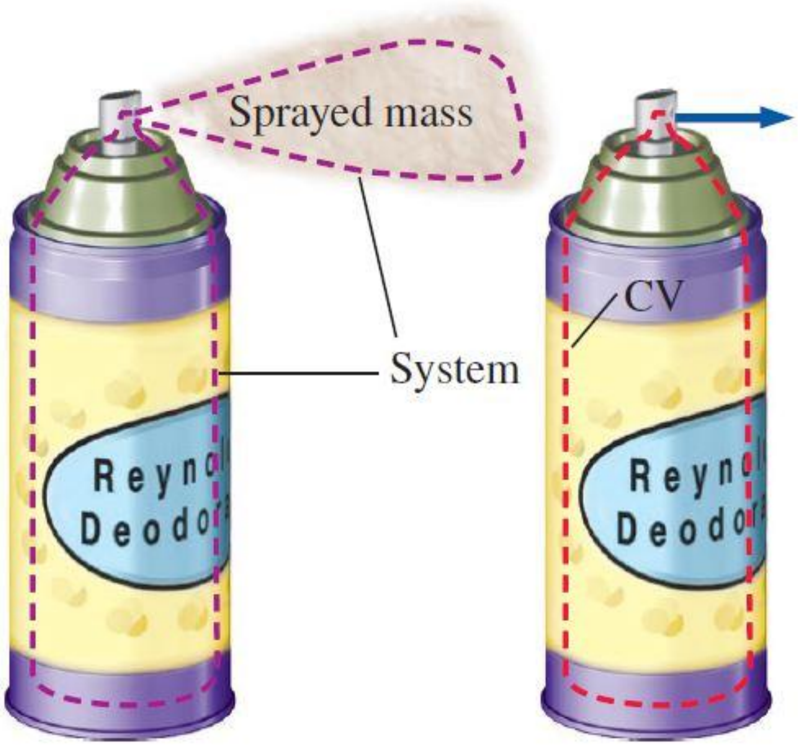
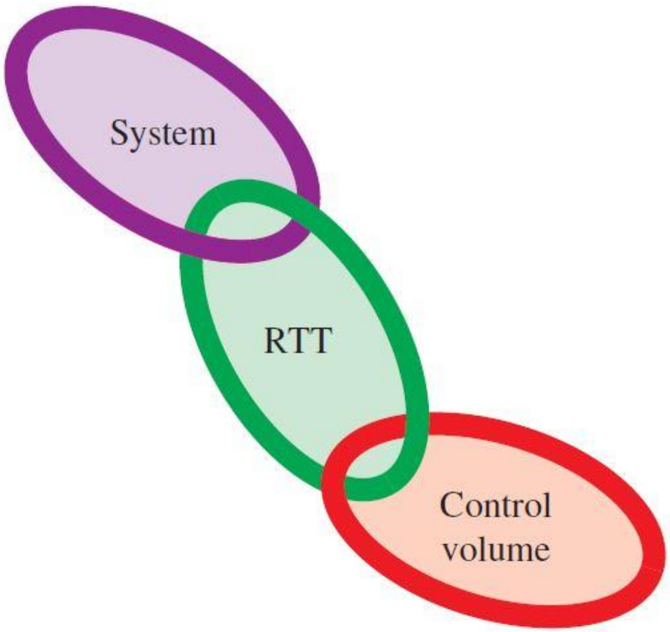


Figure shows a schematic diagram of a control volume showing mass and energy transfers as well as accumulations.

Reynolds Transport Theorem (RTT),
which provides the link between the system and control volume approaches

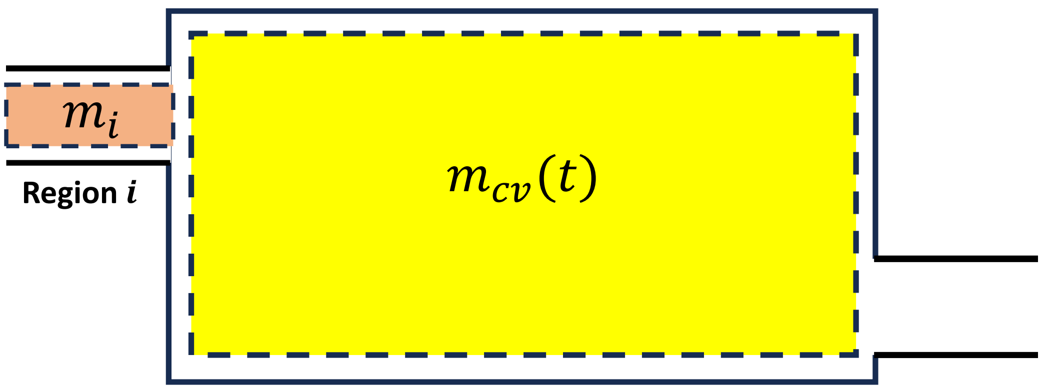


(a) (b)



Conservation of Mass for Control Volume

The conservation of mass principle for a control volume is introduced using the figure which shows a system consisting of a fixed quantity of matter m that occupies different region at time t and a later time $t + \Delta t$

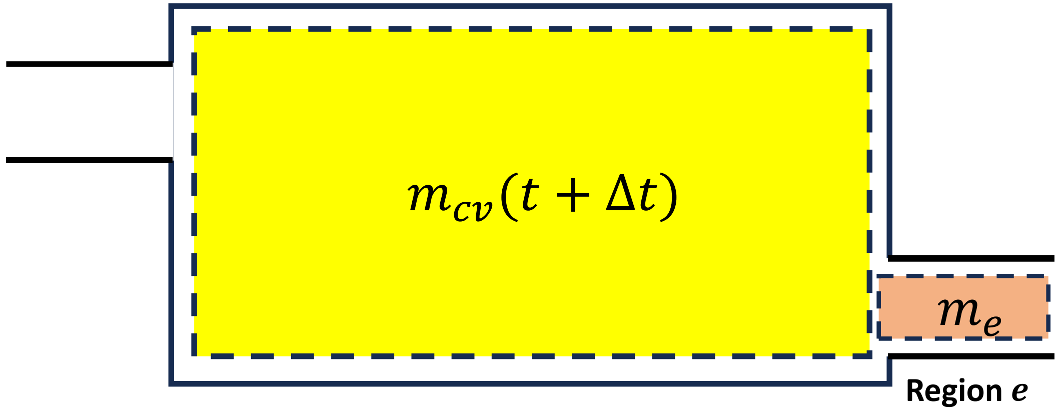


At time t, the amount of mass under consideration

$$m = m_{cv}(t) + m_i$$

Where $m_{cv}(t)$ is the mass contained within the control volume , and m_i is the mass within the small region labelled i adjacent to the control volume as shown in the figure. Let us study the fixed quantity of matter m as time elapsed

In time interval Δt all the mass in the region i enters into the control volume while some of the mass i.e., m_e, initially contained within the control volume exit to fill the region e adjacent to the control volume as shown in the figure



At time t+Δt, the amount of mass under consideration

$$m = m_{cv}(t + \Delta t) + m_e$$

$$m = m_{cv}(t) + m_i = m_{cv}(t + \Delta t) + m_e$$

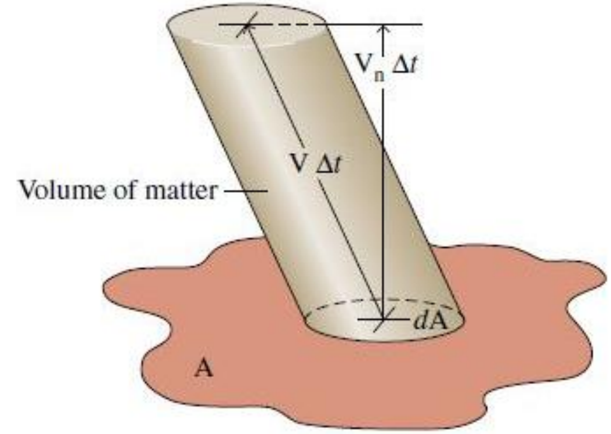
$$m_{cv}(t + \Delta t) - m_{cv}(t) = m_i - m_e$$

$$\lim_{\Delta t \to 0} \frac{m_{cv}(t + \Delta t) - m_{cv}(t)}{\Delta t} = \lim_{\Delta t \to 0} \frac{m_i}{\Delta t} - \lim_{\Delta t \to 0} \frac{m_e}{\Delta t}$$

$$\frac{dm_{cv}}{dt} = \dot{m}_i - \dot{m}_e$$

Amount of mass crossing dA during a time interval Δt

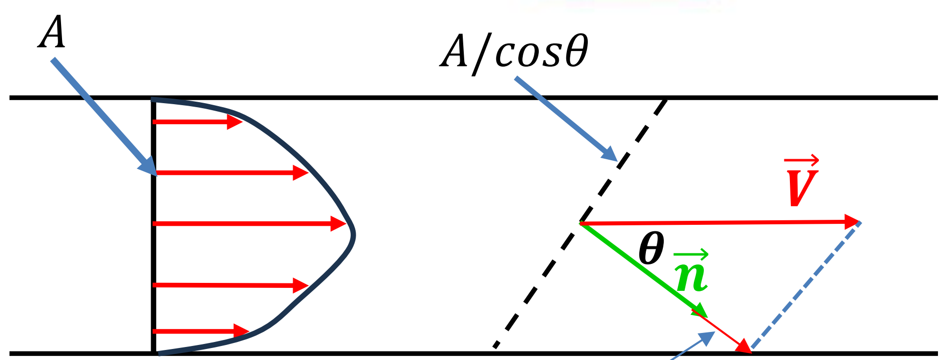
$$\delta m = \rho(\mathrm{V_n}\Delta t)dA$$



Instantaneous rate of mass flow across dA

$$\delta\dot{m} = \rho \mathrm{V_n} dA$$

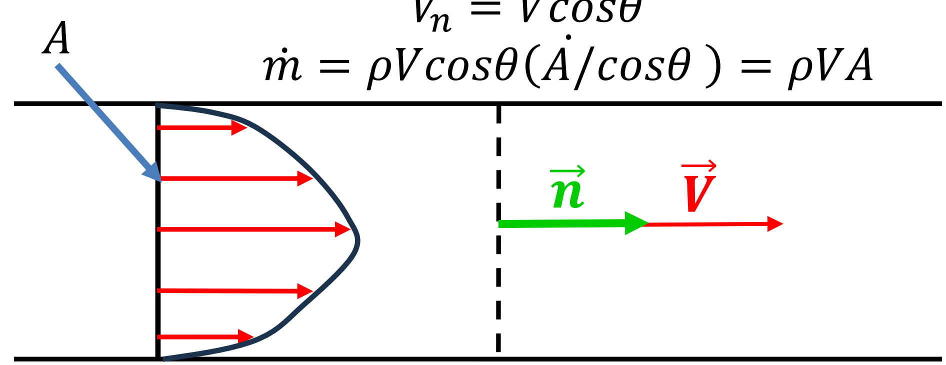
$$\dot{m} = \int_A \rho \mathrm{V_n} dA$$



$$V_n = V\cos\theta$$

$$\dot{m} = \rho V\cos\theta(A/\cos\theta) = \rho VA$$

$$\frac{dm_{cv}}{dt} = \dot{m}_i - \dot{m}_e$$



$$\frac{d}{dt}\int_{CV} \rho dV = \int_{CS} \rho(\vec{V}.\vec{n})dA$$

$$\frac{d}{dt}\int_V \rho dV = \sum_i \int_A \rho V_n dA - \sum_e \int_A \rho V_n dA$$

$$\dot{m} = \rho VA$$

Conservation of mass (continuity equation)

When a flowing flow stream adheres to the following idealisation then the flow is said to be one-dimensional :

- The flow is normal to the boundary at locations where flow enter or leave the control volume
- All intensive properties including velocity and density are uniform with position over each inlet and exit area through which flow enters

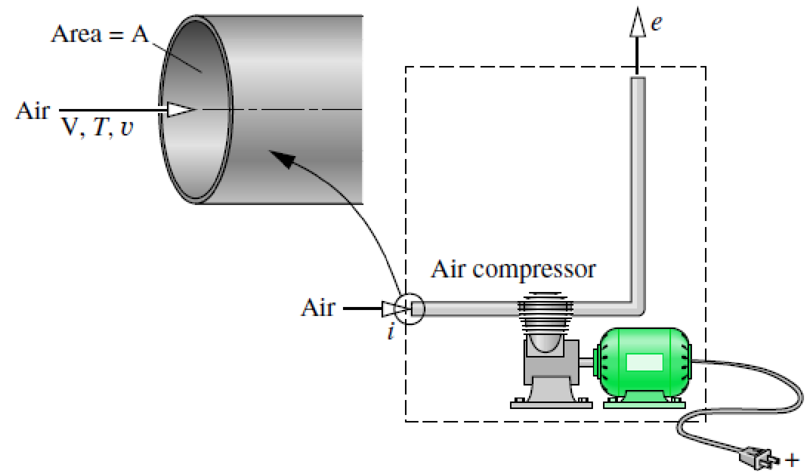


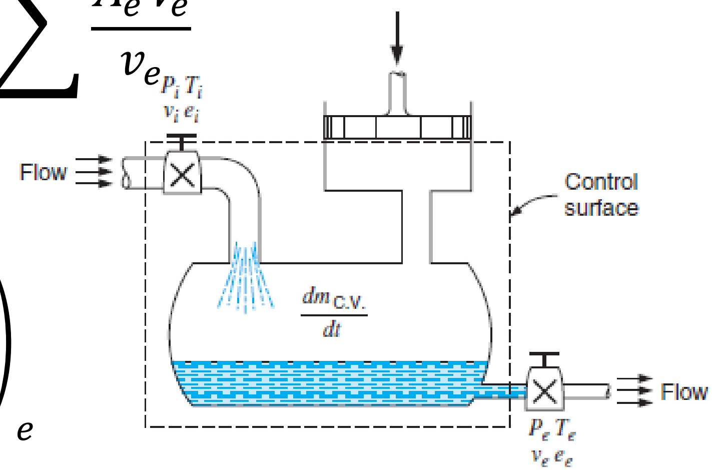
Figure illustrating one-dimensional flow model

When flow is one dimensional: $\dot{m} = \rho A V = \dfrac{AV}{v}$

Continuity equation for 1-D flow: $\dfrac{dm_{cv}}{dt} = \sum \dfrac{A_i V_i}{v_i} - \sum \dfrac{A_e V_e}{v_e}$

Continuity equation in general integral form:

$$\frac{d}{dt}\int_{v} \rho \, dV = \sum_{i} \left(\int_{A} \rho \, V_n dA \right)_i - \sum_{e} \left(\int_{A} \rho \, V_n dA \right)_e$$



Conservation of Total Energy for a Control Volume

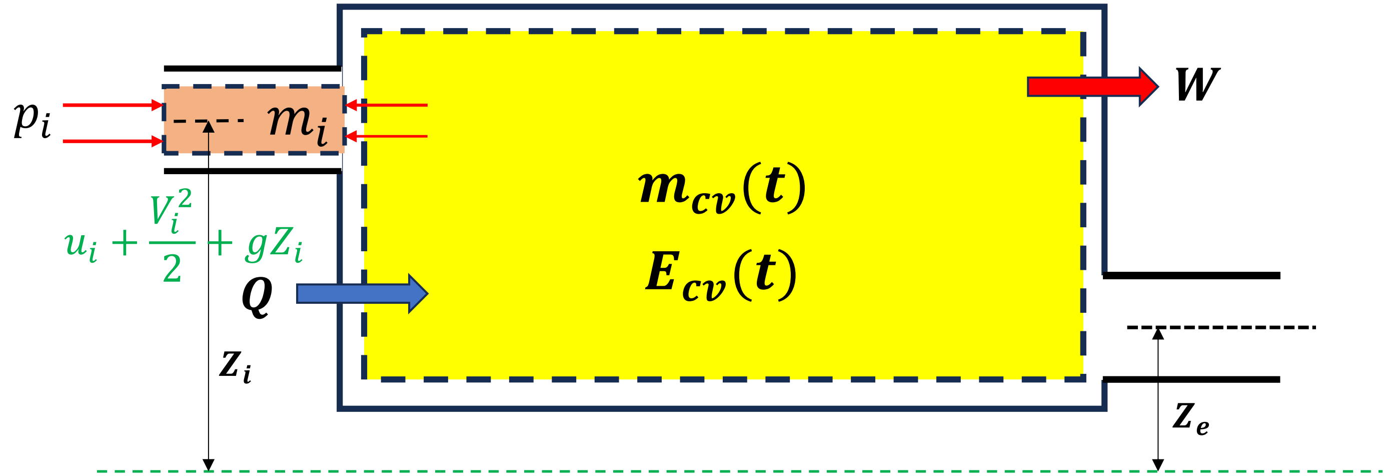
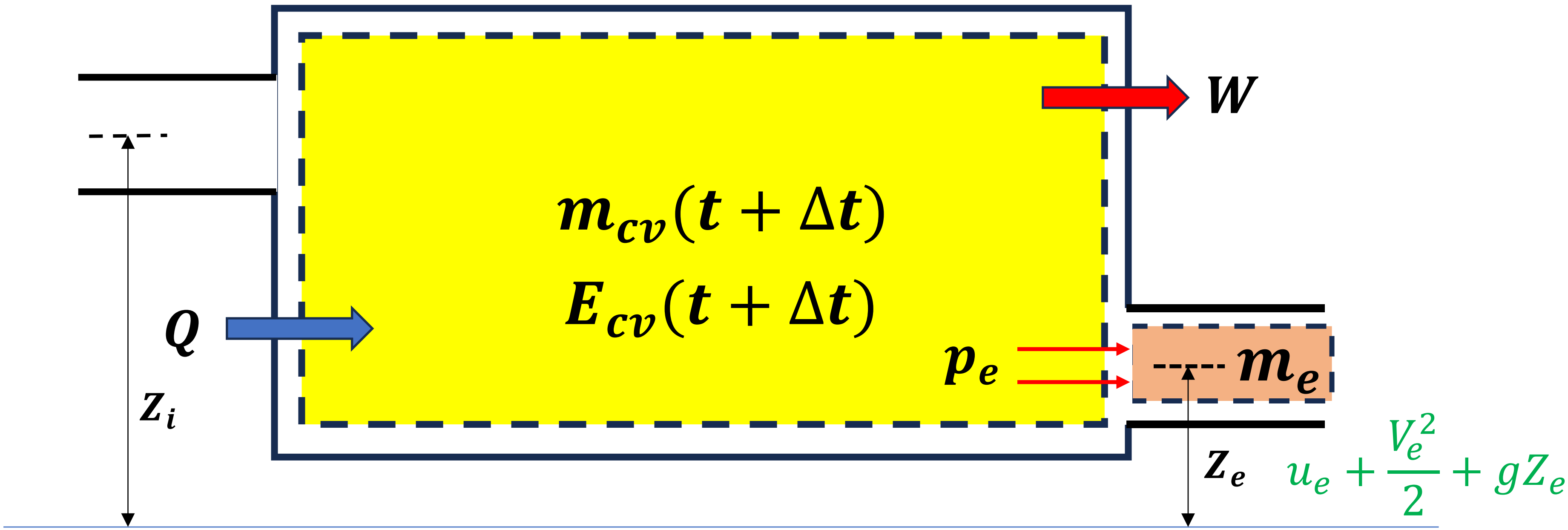


Figure shows a system consisting of fixed quantity of matter m that occupies different regions at time t and time $t + \Delta t$. At time t energy of the system under consideration

$$E(t) = E_{cv}(t) + m_i\left(u_i + \frac{V_i^2}{2} + gZ_i\right)$$

In the time interval Δt all the mass in the region i crosses the control volume boundary while an amount of mass m_e initially in the control volume exits to fill region e, as shown in the figure. During this period there may be energy transfers to and from the system under consideration by heat and work



$$E(t+\Delta t) = E_{cv}(t+\Delta t) + m_e\left(u_e + \frac{V_e^2}{2} + gZ_e\right)$$

Energy balance for the closed system:

$$E(t+\Delta t) - E(t) = Q - W$$

$$E_{cv}(t+\Delta t) + m_e\left(u_e + \frac{V_i^2}{2} + gZ_i\right) - E_{cv}(t) + m_i\left(u_i + \frac{V_i^2}{2} + gZ_i\right) = Q - W$$

$$E_{cv}(t+\Delta t) - E_{cv}(t) = Q - W + m_i\left(u_i + \frac{V_i^2}{2} + gZ_i\right) - m_e\left(u_e + \frac{V_e^2}{2} + gZ_e\right)$$

$$\frac{E_{cv}(t+\Delta t) - E_{cv}(t)}{\Delta t} = \frac{Q}{\Delta t} - \frac{w}{\Delta t} + \frac{m_i\left(u_i + \frac{V_i^2}{2} + gZ_i\right)}{\Delta t} - \frac{m_e\left(u_e + \frac{V_e^2}{2} + gZ_e\right)}{\Delta t}$$

$$\lim_{\Delta t \to 0} \frac{E_{cv}(t+\Delta t) - E_{cv}(t)}{\Delta t} = \frac{dE_{cv}}{dt}$$

$$\lim_{\Delta t \to 0} \frac{Q}{\Delta t} = \dot{Q}$$

$$\lim_{\Delta t \to 0} \frac{W}{\Delta t} = \dot{W}$$

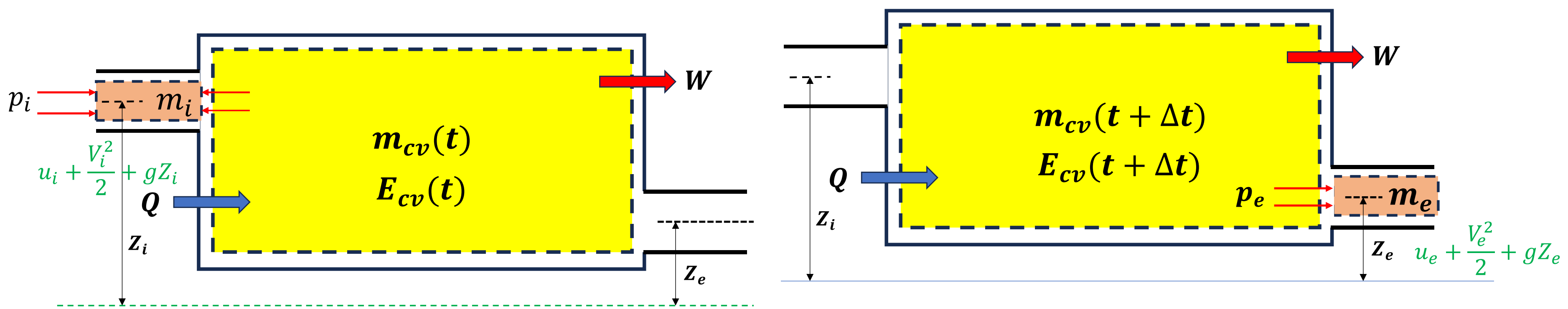
$$\lim_{\Delta t \to 0} \left[\frac{m_i\left(u_i + \frac{V_i^2}{2} + gZ_i\right)}{\Delta t}\right] = \dot{m}_i\left(u_i + \frac{V_i^2}{2} + gZ_i\right)$$

$$\lim_{\Delta t \to 0} \left[\frac{m_e\left(u_e + \frac{V_e^2}{2} + gZ_e\right)}{\Delta t}\right] = \dot{m}_e\left(u_e + \frac{V_e^2}{2} + gZ_e\right)$$

We assumed uniform properties at the inlet and exit of the control volume for each of the mass $\dot{m}_i$ and $\dot{m}_e$ crossing the boundary of the control volume. Consequently in taking the limits above, this corresponds to the assumption 1 D flow at least at the inlet and exit

Evaluating work for a control volume

$\dot{W}$, represents the net rate of energy transfer by work across all portion of the boundary of the control volume. Because work is always done on or by the control volume where matter flows across the boundary

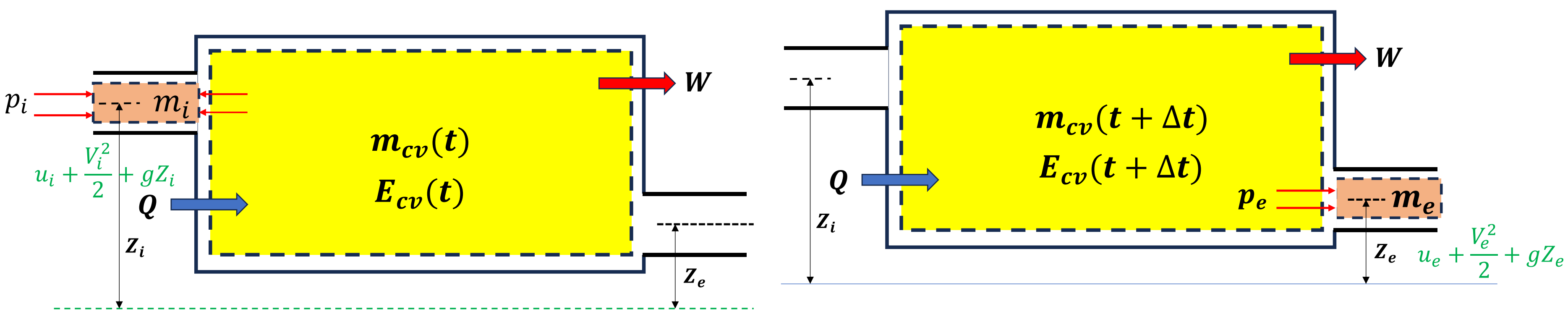


It is convenient to separate the work term $\dot{W}$ into two parts, one is the work associated with the fluid pressure as the mass is introduced at the inlets and removed at the exits. The other contribution, denoted by $\dot{W}_{cv}$, includes all other work effects such as those associated with rotating shaft, displacement of the boundary, electrical, surface tension effects

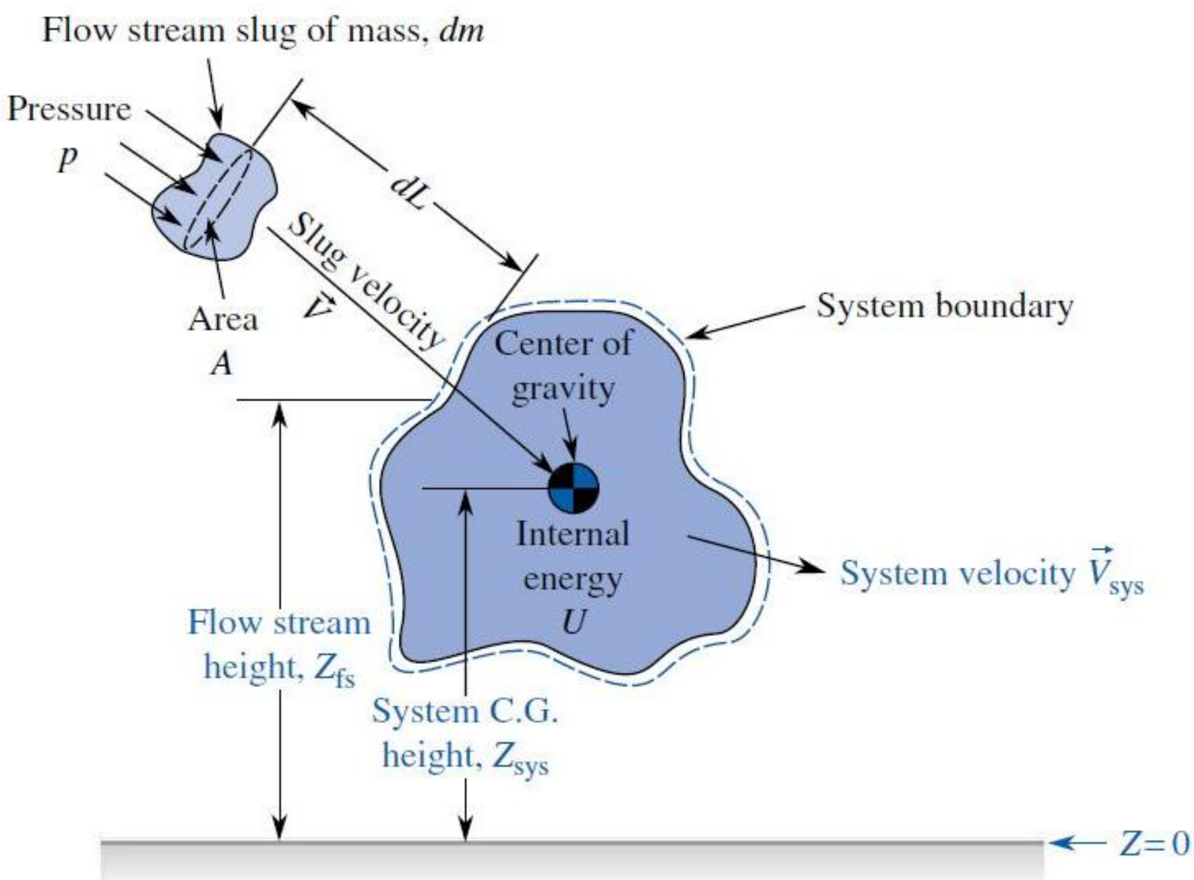
$$\dot{W}_{cv} + \dot{m}_e(p_e v_e) - \dot{m}_i(p_i v_i)$$

$$\frac{dE_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}_i\left(u_i + p_i v_i + \frac{V_i^2}{2} + gZ_i\right) - \dot{m}_e\left(u_e + p_e v_e + \frac{V_e^2}{2} + gZ_e\right)$$

$$\frac{dE_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}_i\left(h_i + \frac{V_i^2}{2} + gZ_i\right) - \dot{m}_e\left(h_e + \frac{V_e^2}{2} + gZ_e\right)$$



The kinetic energy and potential energies appearing in the Storage term (E_{CV}) is the kinetic energy and potential energy of the center of mass of the control volume, however the kinetic & potential energies appearing in the mass advection term are the energies of the working substances at the inlet and exit ports. Hence, they are may or may not be the same.



In a more general form including the local variation of property and considering multiple inlet & exits the energy equation can be written as

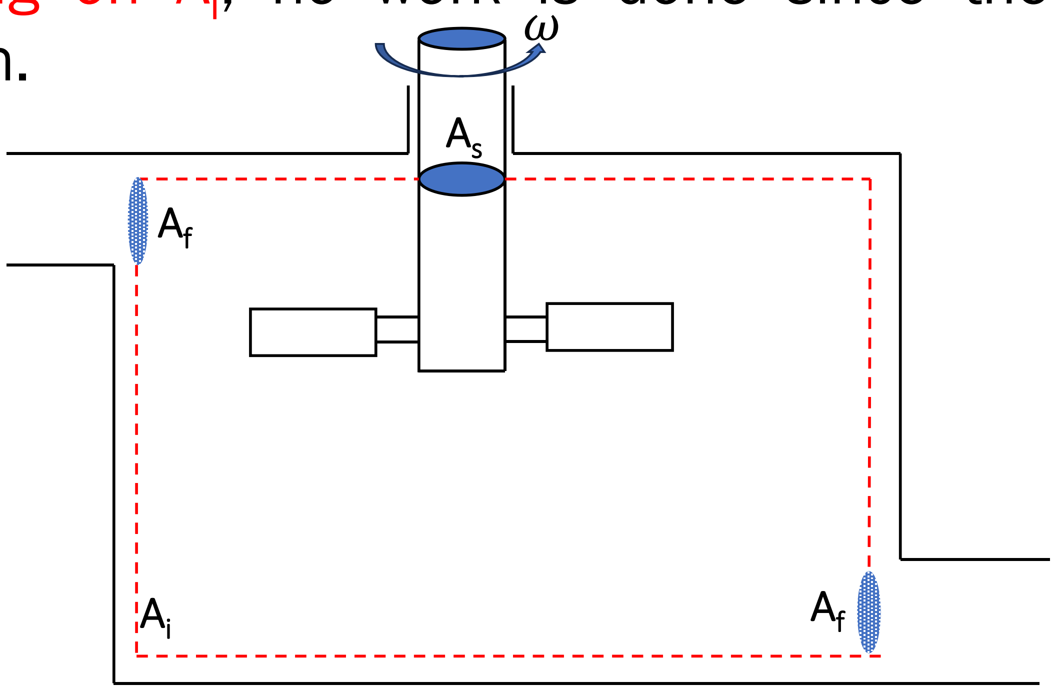
$$E_{CV}(t) = \int_V e\rho dV = \int_V \left(u + \frac{V_{sys}^2}{2} + gz\right)\rho dV$$

$$\frac{d}{dt}\int_V e\rho dV = \dot{Q}_{CV} - \dot{W}_{CV} + \sum_i \left[\int_A \left(h + \frac{V^2}{2} + gz\right)\rho\, V_n dA\right]_i - \sum_e \left[\int_A \left(h + \frac{V^2}{2} + gz\right)\rho\, V_n dA\right]_e$$

Work Done by the Surface Forces **More Insight**

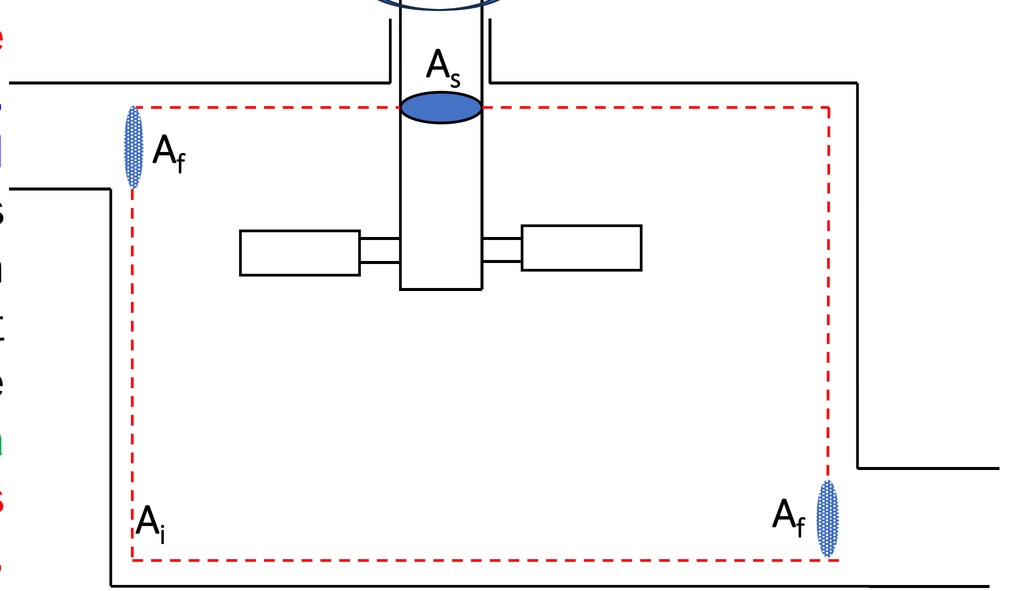
A force does work only if the element on which it acts undergoes a displacement along a line of action of the force. Therefore, not all surface force do work. Figure shows various types of CS elements from the point of view of work done

a) A potion A_i of the control surface may lie adjacent to an impervious solid surface. Even though there are both normal and shear stresses acting on A_i, no work is done since there is no corresponding motion.

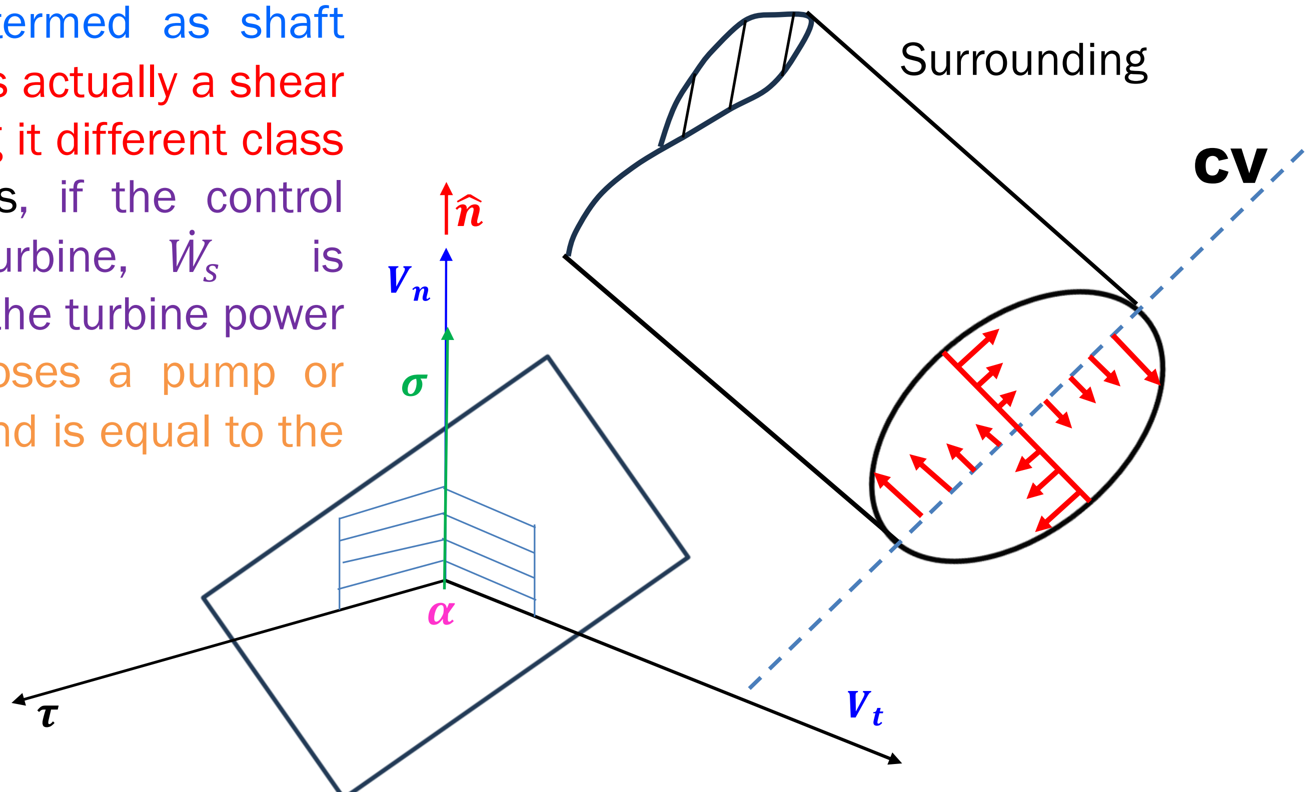


Various types of control surface elements from the point of view of work done

a) A portion A_s of the control surface may cut across a solid body. In general, there will be stresses within the solid body. Generally, two types of such cuts are of engineering interest : one in which the solid is at rest (and does not contribute to the work done) and the other, when the solid rotate, as does a shaft. On rotating shaft, shear stresses act on section As shown in the figure, and do work because of the associated motion. This work is termed as shaft work denoted as $\dot{W}_s$ (it is actually a shear work but we are keeping it different class called shaft wok). Thus, if the control surface encloses a turbine, $\dot{W}_s$ is positive and is equal to the turbine power output. If the CS encloses a pump or blower, $\dot{W}_s$ is negative and is equal to the power input.

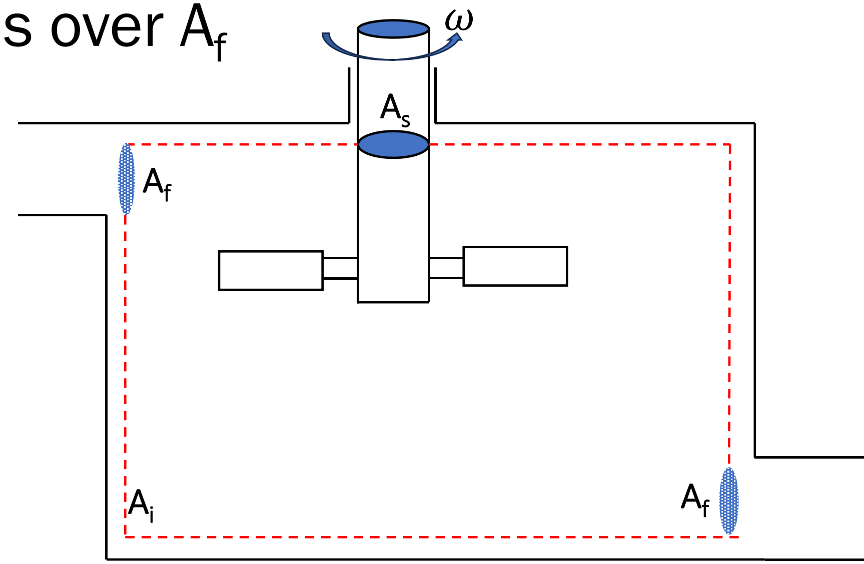


Various types of control surface elements from the point of view of work done



A portion A_f of the CS across which the fluid flows. At such surfaces, both shear and normal stresses do work. Since , in general both normal and tangential components of velocities are present at these location. The total rate of work done by the system at A_f is seen to be

$$-\iint \boldsymbol{V}.(\sigma\hat{\boldsymbol{n}} + \tau\hat{\boldsymbol{t}})dA$$ The integration is over A_f

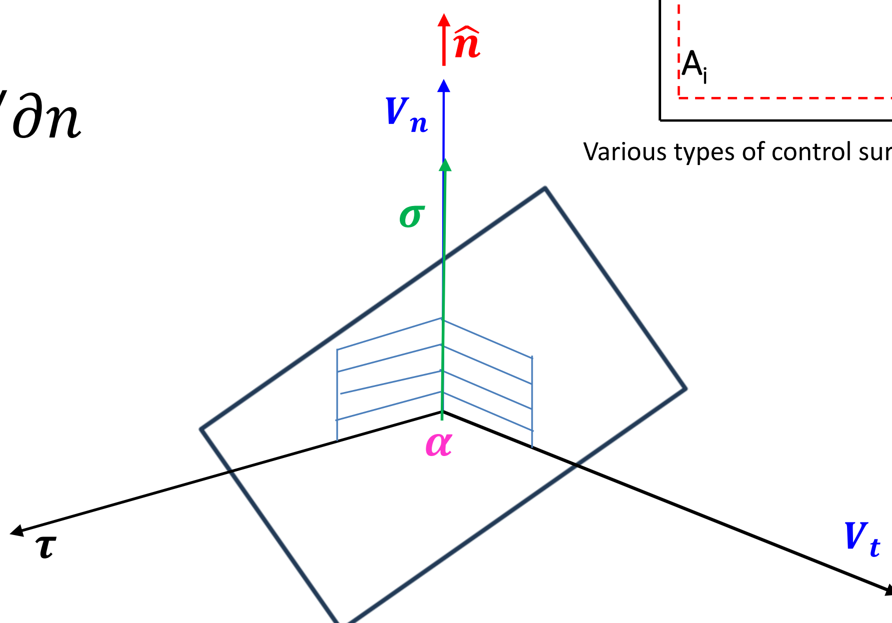


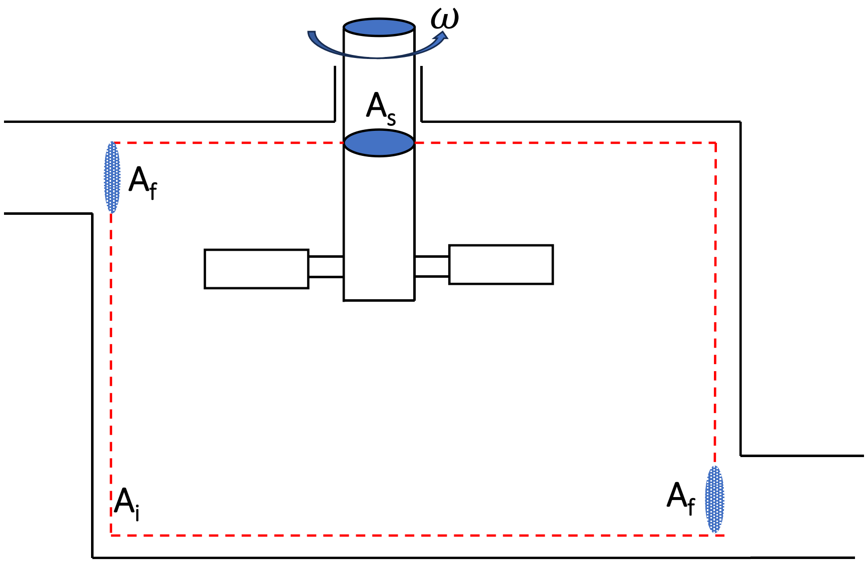
Various types of control surface elements from the point of view of work

$(\sigma\hat{\boldsymbol{n}} + \tau\hat{\boldsymbol{t}})$ is the stress vector acting on the CS.

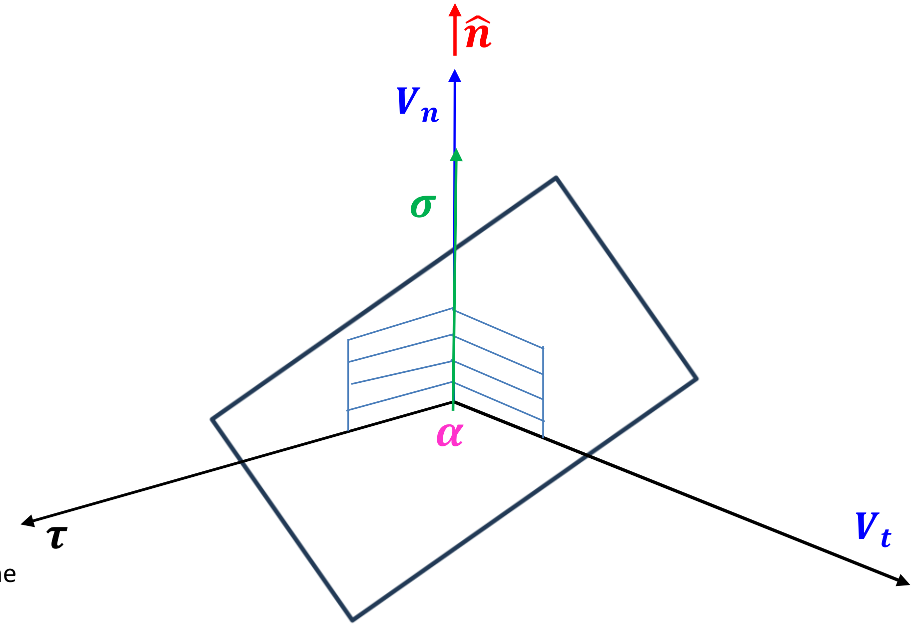
$$\dot{W}_{A_f} = -\iint \tau V_t \cos\alpha dA - \iint \sigma V_n dA$$

$$\sigma = -p + 2\mu\, \partial V_n/\partial n$$





Various types of control surface elements from the point of view of work done



$$\dot{W}_{A_f} = -\iint (\tau V_t \cos\alpha dA + 2\mu\, \partial V_n/\partial n\, V_n) + \iint pV_n dA$$

The 1st integral depends on the viscosity and is termed as viscous work $\dot{W}_\tau$ (in most flows the 2nd term in $\dot{W}_\tau$ is very small)while the 2nd term is called flow work $\dot{W}_f$

$$\dot{W}_{A_f} = -\dot{W}_\tau + \iint p\boldsymbol{V}.dA$$

The Shear work at Ai =0

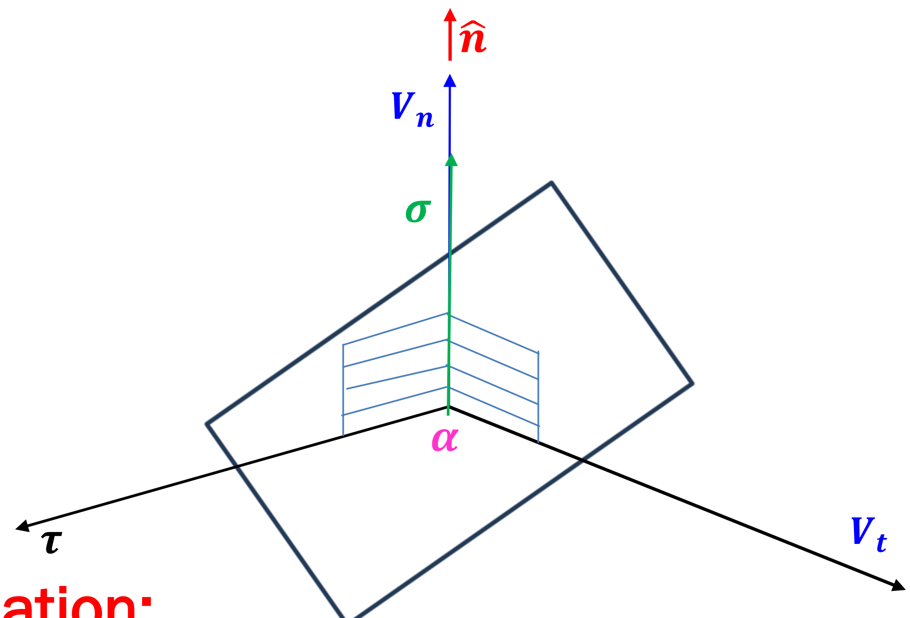
The Control Volume work is

$$\dot{W} = \overset{=0}{\cancel{\dot{W}_{shear}}} + \dot{W}_{shaft} + \dot{W}_{gen} + \dot{W}_{flow}$$

$$\dot{W} = \dot{W}_{shaft} + \dot{W}_{gen} + \int_{A_{in}} Pv d\dot{m} - \int_{A_{out}} Pv d\dot{m}$$

$\dot{\mathrm{W}}_{\mathrm{gen}}$ **includes the PdV, electrical work etc**

The viscous work then vanishes if one of the following conditions holds: the viscosity is negligibly small or the tangential component of the velocity V_t is either zero or it is at right angle to the shear stress everywhere on A_f . With reference to the figure the viscous work drops out when either V is along $\hat{n}$ (i.e, $V_t = 0$), or the angle $\alpha = {}^{\pi}/_{2}$. **We can usually arrange for $\mathbf{V_t}$ and therefore $\mathbf{W_\tau} = \mathbf{0}$ by choosing the CS normal to the velocity vector at the ports.**



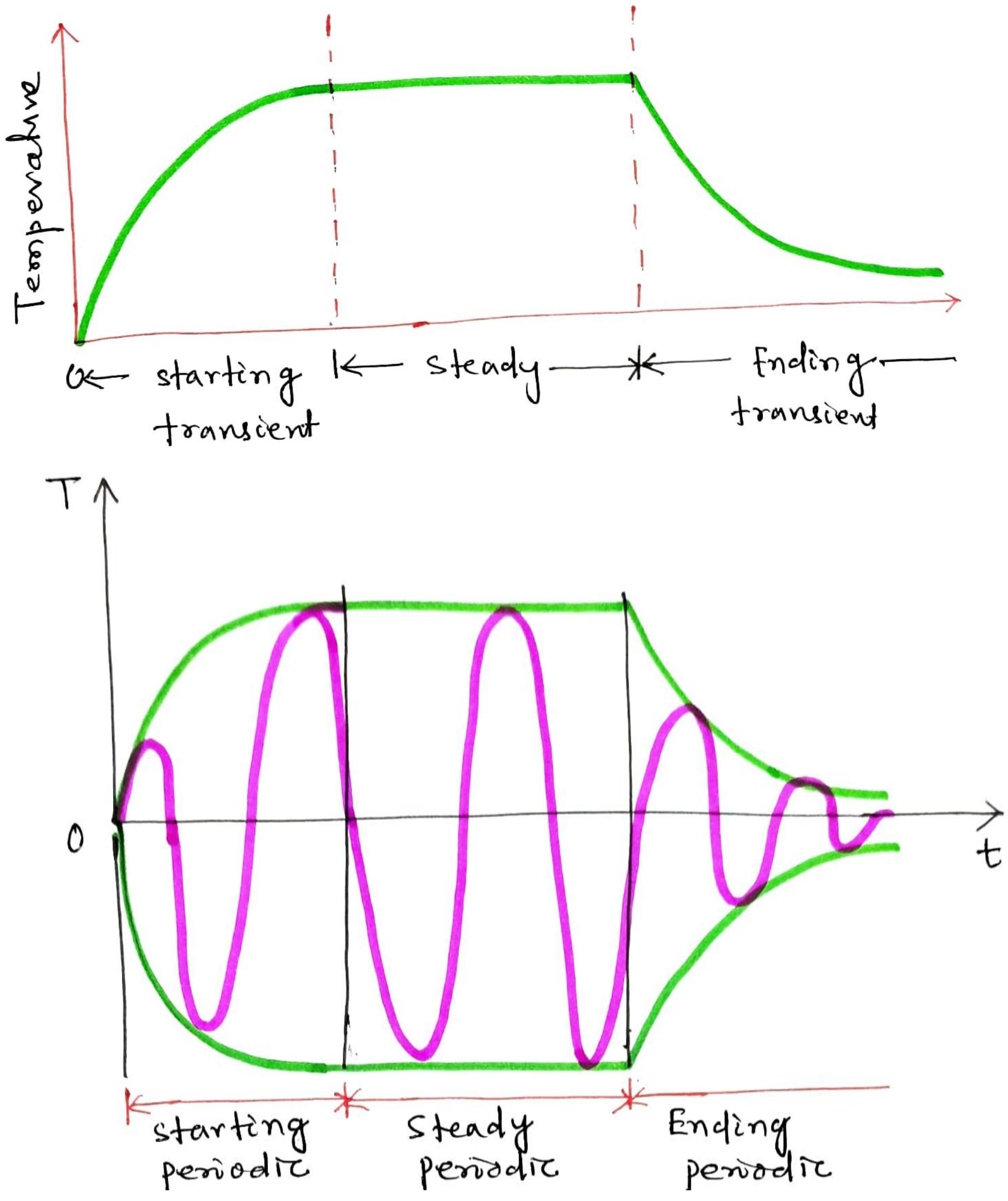
The energy equation:

$$\frac{\partial}{\partial x}\iiint e\rho d\forall + \iint e\rho \boldsymbol{V}.\boldsymbol{dA} = \dot{Q} - \dot{W}_{\boldsymbol{s}} + \dot{W}_{\boldsymbol{\tau}} - \iint p\boldsymbol{V}.\boldsymbol{dA}$$

Steady State Steady Flow Process

The steady state Process:

The first application of the control volume equation will be to develop a suitable analytical model for the long term steady operation of devices such as turbines , compressors, nozzles, boilers etc. This model does not include the short term transients.



Assumptions :

- The control volume does not move relative to the coordinate frame. This means all the velocities measured relative to the coordinate system are also velocities relative to the control surface and there is no work associated with the acceleration of control volume
- The state of the mass at each point of control volume does not vary with time. i.e.

$$\frac{dm_{CV}}{dt} = 0 \Rightarrow \sum \dot{m}_i = \sum \dot{m}_e$$

$$\& \ \frac{dE_{CV}}{dt} = 0$$

- The mass flux and the state of this mass at each discrete area of flow on the control surface does not vary with time.

- The rate at which heat and work cross the control surface remain constant.

$$\dot{Q}_{CV} = C_1;\ \dot{W}_{CV} = C_2$$

Steady flow steady state energy equation:

$$\dot{Q}_{CV} - \dot{W}_{CV} = \dot{m}\left(h_e - h_i + \frac{V_e^2}{2} - \frac{V_i^2}{2} + g(Z_2 - Z_1)\right)$$

$$\delta q - \delta w = (dh + dKE + dPE)$$

$$\delta w = -(vdp + dKE + dPE)$$

$$\delta q + (vdp + dKE + dPE) = (du + pdv + vdp + dKE + dPE)$$

$$\delta q = du + pdv$$

Valid for both simple compressible closed system or steady flow system undergoing frictionless quasi-static process. However, for open system $pdv \neq \delta w$

The steady-state process is often used in the analysis of reciprocating machines such as reciprocating machines such as reciprocating compressors or engines. In this case the rate of flow , which may be actually be pulsating , is considered to be average rate of flow for an integral number of cycles, A similar assumption is made regarding the properties of the fluid flowing across the control surface and the heat transfer and work crossing the control surface. It is also assumed that for integral number of cycles the reciprocating device undergoes, the energy and mass within the control volume do not change